

LIFE, ENVIRONMENT AND HEALTH

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3.1 INTRODUCTION

The Indian Knowledge System (IKS) is a treasure trove of wisdom that encompasses a broad spectrum of human understanding, particularly in the realms of life, the environment, and health. Rooted in ancient philosophical and spiritual traditions, IKS offers profound insights into the interconnectedness of human life with the natural world. For centuries, Indian scholars and practitioners have sought to understand life in a holistic manner, where every aspect of existence is intricately linked to the other. This perspective is seen not only in their study of physical health but also in their broader approach to life sciences, including plant science, anatomy, physiology, and agriculture.

The traditional Indian view of life sciences emphasizes balance and harmony. Health was seen as more than just the absence of disease; it was a state of equilibrium between the body, mind, and the environment. The ancient medical systems, such as Ayurveda and Siddha, emphasize the balance of bodily elements (doshas) and the importance of aligning oneself with the rhythms of nature. This holistic approach was complemented by practices of meditation, yoga, and spiritual reflection, all of which contributed to overall well-being.

Indian knowledge also extended to the natural world, with a deep understanding of plant life, ecosystems, and environmental sustainability. The ancient Indians recognized the importance of biodiversity and the need for humans to live in harmony with nature. Agricultural practices in India, such as crop rotation, organic farming, and rainwater harvesting, were not only based on careful observation of the environment but also carried spiritual significance. India's traditional wisdom concerning agriculture focused on maintaining soil fertility, protecting water sources, and ensuring sustainable food production.

The Indian approach to environmental stewardship was deeply embedded in its spiritual and cultural practices. Sacred groves, where no trees were cut and no animals were harmed, were a common feature across India. These environmental conservation practices were tied to religious beliefs, where certain trees and animals were considered sacred and were revered. The ancient Indians believed that protecting nature was a moral and spiritual responsibility.

This profound respect for life and the environment also found expression in India's rich literature and scriptures. Texts like the Vedas, Upanishads, and the Puranas provide countless examples of how to live in harmony with nature. They often featured stories and teachings that emphasized the interdependence of all living beings and the need for ecological balance. Whether it was through the worship of natural forces, the reverence for rivers like the Ganga, or the practice of sustainable farming, India's knowledge systems promoted a life that was integrated with nature.

In modern times, as the world faces environmental challenges, the relevance of these ancient practices is increasingly being acknowledged.

The Indian Knowledge System, with its deep emphasis on sustainability, health, and the environment, continues to inspire practices of eco-conscious living, traditional medicine, and ecological preservation. By looking back at these ancient teachings, we can find solutions that are both effective and sustainable for the modern world.

In this unit, we will explore how ancient Indian knowledge contributed to life sciences, health, and environmental sustainability, and how these traditions can help us build a more harmonious and sustainable future. Through understanding traditional Indian perspectives on life, the environment, and health, we can discover a timeless wisdom that continues to be relevant in today's world.

3.2 APPROACHES TO LIFE AND HEALTH : TRADITIONAL INDIAN PERSPECTIVES

In ancient India, health and life sciences were approached in a holistic way, where the connection between the human body, mind, and environment was recognized and valued. The traditional Indian understanding of life was rooted in deep spiritual and philosophical ideas, which emphasized the balance of nature and the interdependence of all living beings. The study of life sciences in India was broad and inclusive, covering plant science, human anatomy and physiology, and even agriculture. These areas were not seen as isolated from each other; rather, they were all part of a larger system that focused on maintaining harmony with nature and ensuring overall well-being.

3.2.1 Plant Science in Ancient India

In ancient India, plants were considered vital not just for food, but for their medicinal and spiritual significance as well. The knowledge of plants was not only practical but also rooted in philosophy and religion. Indian scholars and sages studied plants extensively, identifying their medicinal properties and understanding how they could be used to maintain health and cure diseases.

- **Ayurveda and plants:** One of the most prominent systems of medicine in India, Ayurveda, has always placed a special emphasis on plants. Many plants were used for their healing properties, and the study of herbs was a core part of Ayurvedic practice. Texts like the *Charaka Samhita* and *Sushruta Samhita* list numerous herbs and their

benefits. For example, *Tulsi* (holy basil) is considered sacred and is used to treat a variety of ailments, while *Neem* is known for its purifying properties.

- **Botanical knowledge:** Indian botanists were also well aware of the importance of plants in ecological balance. They understood that plants not only provided food and medicine but were essential for the survival of all other life forms, including humans. The ancient Indian texts often describe plants as interconnected with the well-being of the earth and the universe.

3.2.2 Anatomy and Physiology in Traditional Indian Systems

The study of the human body in traditional Indian medicine was highly advanced and detailed. The ancient Indians believed that the human body was a microcosm of the universe, and that maintaining balance within the body was key to health.

- **Ayurvedic view of the body:** According to Ayurveda, the human body is made up of five elements—earth, water, fire, air, and space. These elements form the *doshas* (bodily energies): *Vata* (air), *Pitta* (fire), and *Kapha* (earth and water). Good health is believed to result from a balance between these doshas, while disease arises from their imbalance. Ayurvedic practitioners were skilled in diagnosing and treating these imbalances through diet, lifestyle changes, herbs, and therapies like massages and detoxification.
- **Human anatomy:** Ancient Indian texts like the *Sushruta Samhita* describe human anatomy in great detail. Sushruta, known as the father of surgery, is credited with writing about the structure of the body, the functions of organs, and surgical techniques. He also developed methods for the treatment of wounds, fractures, and even plastic surgery, demonstrating a high level of medical knowledge.
- **Yoga and physiology:** Another key aspect of traditional Indian approaches to health is the practice of yoga, which focuses on the integration of the body, mind, and spirit. Yoga techniques involve physical postures (*asanas*) that help in maintaining the health of the

body, alongside breathing exercises (pranayama) to balance the mind. These practices promote circulation, flexibility, and overall health, emphasizing that physical well-being is connected to mental clarity.

3.2.3 Agriculture and its Role in Health

Agriculture played a central role in ancient Indian life, and the approach to farming was closely linked to both health and the environment. Traditional agricultural practices in India were based on a deep understanding of nature and the rhythms of the earth.

- **Sustainable farming:** Indian farmers traditionally followed sustainable farming practices that aimed to preserve the fertility of the soil and protect water resources. Techniques like crop rotation, composting, and the use of organic fertilizers helped maintain soil health. These methods also prevented the overuse of natural resources, ensuring that the land could continue to provide food for generations.
- **The connection between agriculture and health:** In India, agriculture was seen as not only a means of producing food but also as a way to promote health and well-being. The concept of *Sattvic* (pure) food in Ayurveda, which is vital for good health, emphasized food grown naturally and organically. This means that the health of the soil and the environment was directly linked to the health of the people consuming the food grown on it.
- **Holistic view of food:** In Indian traditions, food was considered a form of medicine. Different foods were believed to have different effects on the body based on their properties and the balance of the doshas. For example, fresh fruits and vegetables were seen as cooling and nourishing, while spicy foods were considered stimulating and heating. The traditional Indian diet was designed not only to provide nutrition but also to maintain harmony within the body.

In traditional Indian perspectives, life and health are seen as interconnected aspects of a larger cosmic system. Health is not merely the absence of illness but a state of harmony between the body, mind,

and the environment. The Indian Knowledge System, especially through practices like Ayurveda, yoga, and sustainable agriculture, provide a comprehensive approach to maintaining good health and a deep understanding of the relationship between humans and nature. By looking back at these ancient wisdoms, we can find valuable insights to improve our health and well-being in today's world.

3.3 ENVIRONMENTAL AWARENESS: INDIA'S TRADITIONAL EMPHASIS ON ECOLOGY AND ENVIRONMENTAL SUSTAINABILITY

India has a long history of environmental awareness and sustainability, rooted in its cultural, religious, and philosophical traditions. The people of ancient India understood that human life is deeply connected to nature, and they viewed the environment as sacred. Over thousands of years, India developed a range of practices and principles that helped in maintaining ecological balance, conserving natural resources, and promoting sustainability. These practices are still relevant today and offer valuable lessons for protecting the environment.

3.3.1 The Sacred Relationship with Nature

In traditional Indian culture, nature was not seen as something separate from human life, but as an integral part of it. Rivers, trees, mountains, and animals were revered as sacred, and many Hindu deities were associated with natural elements.

- **Sacred rivers and forests:** Rivers like the Ganges, Yamuna, and Saraswati have been considered sacred in Hinduism, and their importance goes beyond spiritual beliefs. These rivers provided water for agriculture, drinking, and sanitation. People living along these rivers practiced rituals and offered prayers to honor these natural resources, ensuring they were not exploited recklessly.
- **Sacred groves:** In many parts of India, groves of trees were protected and treated with reverence. These sacred groves were considered the abodes of gods and were preserved by local communities as places of worship and conservation. The people understood that trees were vital for sustaining life by providing oxygen, preventing soil erosion, and maintaining biodiversity.

3.3.2 Principles of Sustainability in Ancient Practices

The ancient Indian way of life was deeply rooted in the principles of sustainability. Sustainability was not just about using resources wisely, but also about ensuring that nature and human life were in harmony.

- **Ahimsa (non-violence):** One of the core principles in Indian philosophy, *ahimsa*, or non-violence, extended to all living beings. This principle encouraged people to live in a way that caused the least harm to nature and animals. This helped protect wildlife and natural habitats from over-exploitation.
- **Conservation of forests:** In ancient India, forests were seen as essential for the balance of life. Various texts, like the *Arthashastra*, advocated for the protection of forests and wildlife. Communities living near forests followed sustainable practices like selective tree cutting and non-destructive hunting.
- **Water conservation:** Water was seen as a sacred resource, and many traditional Indian practices focused on conserving and managing water efficiently. Ancient irrigation systems, such as step-wells, ponds, and rainwater harvesting systems, were built to store and distribute water. These practices ensured that water was available during dry periods and helped avoid wastage.

3.3.3 Sustainable Agriculture

Agriculture has always been central to Indian life, and traditional farming practices were designed to be sustainable and eco-friendly.

- **Crop rotation and mixed farming:** Ancient Indian farmers practiced crop rotation to maintain soil fertility. By planting different crops in succession, they ensured that the soil did not lose its nutrients. They also practiced mixed farming, where both crops and animals were raised together, creating a balance between plant and animal life.
- **Organic farming:** The traditional Indian approach to agriculture avoided the use of chemicals and synthetic fertilizers. Instead, organic farming methods were used, such as composting, green manuring, and using natural pesticides made from plants like neem. This helped preserve the health of the soil and prevented harmful effects on the environment.

- **Sowing according to seasons:** Ancient farmers in India had a deep understanding of the seasons and weather patterns. They sowed crops according to the time of year and the climate conditions, ensuring that crops were grown at the right time and that the land could regenerate naturally between planting cycles.

3.3.4 Conservation of Wildlife

Ancient Indian practices also focused on the protection of wildlife. The concept of *ahimsa* extended to animals, and people were encouraged to protect animals and prevent their harm.

- **Wildlife sanctuaries:** Some ancient Indian rulers established protected areas or wildlife sanctuaries. For example, King Ashoka of the Maurya Dynasty declared several forests as protected areas where hunting was prohibited. These were some of the earliest forms of wildlife conservation in the world.
- **Animal worship:** Many animals were considered sacred and were worshipped as manifestations of gods and goddesses. The cow, for example, was revered and protected, and the killing of cows was prohibited in many regions. This led to the conservation of certain species and the development of rules that restricted the exploitation of animals.

3.3.5 Environmental Stewardship through Festivals and Rituals

In India, many festivals and rituals celebrate nature and promote environmental consciousness.

- **The festival of trees - vata savitri vrat and van mahotsav:** In these festivals, people worship trees and plants, showing respect for their role in sustaining life. During Van Mahotsav, which is observed during the monsoon season, tree planting ceremonies are held, where communities come together to plant and protect trees.
- **The festival of water - gangasagar mela:** The Gangasagar Mela, which celebrates the confluence of the Ganges River with the Bay of Bengal, is another example of the reverence for water bodies. Pilgrims gather to bathe in the river, and the event includes efforts to clean and protect the river from pollution.

3.3.6 Ayurvedic Practices and Environmental Harmony

Ayurveda, the traditional Indian system of medicine, also reflects a deep connection between health and the environment. Ayurveda encourages the use of natural, plant-based medicines, many of which come from local plants and herbs that are cultivated in harmony with nature.

- **Holistic health:** Ayurvedic practitioners emphasize the balance between the body and the environment. The seasonal changes, the food we eat, and the climate all affect our health, and Ayurveda teaches how to live in balance with nature to promote overall well-being.
- **Herbal conservation:** Many of the medicinal plants used in Ayurveda are rare and require protection. Traditional practices focused on the cultivation of these plants in sustainable ways, ensuring that they did not become endangered.

3.3.7 Modern Relevance of Traditional Environmental Practices

Today, as the world faces environmental crises such as climate change, pollution, and loss of biodiversity, the traditional environmental practices of India offer valuable lessons in sustainability. Concepts such as water conservation, organic farming, and the protection of natural resources are more relevant than ever.

India's traditional ecological knowledge provides a framework for living in harmony with the earth, and its principles can guide modern efforts to create a more sustainable and environmentally conscious world.

India's traditional environmental awareness, rooted in religious beliefs, cultural practices, and sustainable living, offers timeless wisdom for ecological conservation. The connection between human life and nature was seen as sacred, and protecting the environment was considered a moral responsibility. By understanding and adopting these ancient practices, we can work towards a future where humans live in balance with the earth, ensuring the survival and flourishing of all life forms.

Exercise**(A) Multiple Choice Questions (MCQs):**

1. What is the core concept of health in the Indian Knowledge System (IKS)?
(a) The absence of disease
(b) The balance between the body, mind, and environment
(c) Physical fitness
(d) Mental clarity
2. Which traditional Indian medical system emphasizes the balance of bodily elements (doshas)?
(a) Unani
(b) Ayurveda
(c) Homeopathy
(d) Naturopathy
3. What is one of the primary practices of sustainability in traditional Indian agriculture?
(a) Monocropping
(b) Crop rotation
(c) Use of chemical pesticides
(d) Large-scale mechanization
4. What does the term "Sattvic food" in Ayurveda refer to?
(a) Processed and packaged food
(b) Food grown naturally and organically
(c) Food rich in sugar
(d) Fast food
5. Which river is considered sacred in Hinduism and plays a crucial role in environmental consciousness in India?
(a) Nile
(c) Ganga
(b) Amazon
(d) Yangtze
6. What principle encourages the least harm to nature and animals in Indian philosophy?
(a) Karma
(b) Ahimsa
(c) Dharma
(d) Moksha
7. Which traditional practice in India emphasizes the cultivation of medicinal plants for sustainability?
(a) Organic farming
(d) Both (b) and (c)
(b) Ayurveda
(c) Sacred groves

8. What was the purpose of ancient irrigation systems like step-wells and ponds in India?
- (a) To increase agricultural productivity
 - (b) To promote tourism
 - (c) To store and distribute water**
 - (d) To water plants in urban areas
9. What was the role of sacred groves in ancient India?
- (a) To serve as places of worship and environmental conservation**
 - (b) To serve as agricultural fields
 - (c) To serve as military bases
 - (d) To serve as market places
10. Which ancient Indian ruler established protected forests to conserve wild life?
- (a) King Ashoka**
 - (b) Emperor Akbar
 - (c) King Vikramaditya
 - (d) King Harsha

(B) Short Answer Questions:**(1 Mark)**

1. What is Ayurveda primarily focused on?
2. Name one sacred river in India.
3. What does "Ahimsa" mean in Indian philosophy?
4. Which system of ancient Indian medicine is associated with the balance of doshas?
5. What is the significance of sacred groves in India?

(C) Short Answer Questions:**(2 Marks)**

1. Explain the connection between the Indian Knowledge System and sustainable agricultural practices.
2. How did ancient Indian rulers contribute to the conservation of wild life?

3. What role did sacred groves play in environmental conservation in India?
4. Describe the traditional Indian approach to water conservation and management.
5. How does Ayurveda integrate the concept of balance in health with environmental sustainability?

(D) Long Answer Questions:**(3 Marks)**

1. Describe the Ayurvedic understanding of the human body and its elements.
2. How does the Indian Knowledge System approach the interdependence between humans and nature?
3. Discuss the significance of plants in Indian life sciences, including their medicinal and ecological roles.
4. Explain the concept of Sattvic food in Ayurveda and its importance for health.
5. How does the practice of yoga contribute to both individual health and environmental awareness in traditional Indian thought?

(E) Long Answer Questions:**(6 Marks)**

1. Discuss the traditional Indian approach to environmental sustainability and its relevance today.
2. Analyze how India's traditional medical systems, such as Ayurveda, emphasize the relationship between health and the environment.
3. Describe the principles of sustainability in ancient Indian agricultural practices and how they contributed to the long-term health of the land.
4. Explain the concept of Ahimsa in Indian philosophy and its impact on environmental and wildlife conservation.
5. Discuss the role of sacred rivers and forests in ancient Indian society and their connection to both spirituality and environmental preservation.

Solved Previous Question Paper

B.Sc. & B.Sc. (ECS) I : Sem I
Introduction to Indian Knowledge System (IKS)
AY 2024-25 University Assessment (UA)
Day, Date: Friday, 03-01-2025

Total Marks: 30

Time:

Instructions:

- (1) All questions are compulsory.
- (2) Figures to right indicate full marks.

Q.1 Choose the correct alternative (MCQs): (6 Marks)

1. The book "A History of Hindu Chemistry from Earliest Times to Medieval of 16th century" is written by
(a) J.C. Bose
(b) Tagore
(c) **Acharya Prafulla Chandra Roy**
(d) Netaji
2. The **Vimana** refers largely and exclusively to the flying vehicles used by Ashvins (vedic gods of medicine)
(a) **Rigveda** (b) Yajurveda
(c) Atharvaveda (d) Samaveda
3. **Mahatspota** means
(a) Big spot (b) **Big bang**
(c) Big ball (d) Big disaster
4. **Bijaganita** describes zero and infinity.
(a) **Bhaskaracharya** (b) Chanakya
(c) Chandragupta (d) Vikramacharya
5. The **Vedas** mentions protecting ecology and the environment
(a) **Rigveda** (b) Yajurveda
(c) Atharvaveda (d) Samaveda
6. Chanakya was an alumnus of
(a) **Takṣaśilā** (b) Vikramshila
(c) Nālandā (d) Vallabhi

Q.2 Answer the following (Any Three): (6 Marks)

(Each question carries 2 marks)

(A) Write about the concept of pi (π):

Ans. Pi (π) is a mathematical constant that represents the ratio of the circumference of a circle to its diameter. Ancient Indian mathematician Aryabhata, in his work *Aryabhatiya*, gave an approximation of pi as 3.1416, which was remarkably accurate for its time. His work laid the foundation for the study of geometry and trigonometry in both Indian and global mathematics. Later, Indian scholars like Bhaskara II refined its calculation to enhance mathematical precision.

(B) Define ecology.

Ans. Ecology is the scientific study of interactions between living organisms and their physical environment. It focuses on maintaining balance and sustainability in ecosystems. In ancient Indian traditions, ecology emphasized harmony with nature, promoting practices like sustainable agriculture, afforestation, and conservation of resources.

(C) Define philosophy.

Ans. Philosophy is the study of fundamental questions about life, existence, knowledge, ethics, and the universe. It uses reasoning, logic, and critical thinking to explore the nature of reality and human purpose. Indian philosophy, in particular, is deeply intertwined with spiritual concepts like moksha (liberation) and dharma (righteousness).

(D) Define physiology.

Ans. Physiology is the branch of biology that studies the normal functions of living organisms and their parts. It explores how organs, systems, and cells interact to sustain life. Ancient Indian medicine, such as Ayurveda, incorporates physiological concepts like *doshas* (bodily energies) to explain health and disease.

(E) Vedic Mathematics.

Ans. Vedic Mathematics is an ancient Indian system of mathematical techniques derived from Vedic texts. It simplifies complex calculations through mental strategies and 16 sutras (principles). These techniques are applicable to arithmetic, algebra, and geometry, and have greatly influenced global mathematical practices.

Q.3 Answer the following (Any Two):**(6 Marks)****(Each question carries 3 marks)****(A) Describe Indian concepts related to the atom.**

Ans. The concept of the atom (*Anu*) was introduced by the ancient Indian philosopher Kanada in his *Vaisheshika Sutra*. He proposed that all matter is made up of indivisible particles called atoms. These atoms are eternal, indivisible, and cannot be perceived by the naked eye. They combine in various ways to form complex substances.

Kanada also identified different properties of atoms, such as motion and interaction. According to him, atoms move continuously, and their combination creates the material world. He even described atomic theory in terms of causes and effects,

stating that the physical world emerges from the combination of atoms.

This early atomic theory predates similar ideas in Greek philosophy by several centuries and is a testament to the advanced nature of Indian scientific thought. Kanada's concepts were not limited to theoretical discussions; they influenced later scholars, laying a foundation for scientific inquiry in India.

(B) Explain Indian innovations in daily-life chemistry.

Ans. Ancient Indian chemists made remarkable innovations that directly impacted daily life, particularly in the fields of metallurgy, dye production, glass making, and medicine.

One of the most notable contributions was the creation of **Wootz steel**, a rust-resistant and highly durable material used in weapons and tools. Wootz steel was so advanced that it became highly sought after in the Middle East and Europe.

Indian artisans also perfected the art of creating vibrant natural dyes, such as **indigo** and **madder**, which were used in textiles and traded globally. They mastered the techniques of dyeing and printing fabrics, creating patterns that remain significant in Indian art and culture.

In addition, Indian chemists were skilled in **glass and pottery production**. They developed methods for producing high-quality glass beads and ceramic objects, many of which were used in daily life. The use of **lime-based cements** for construction is another testament to their advanced chemical knowledge.

Furthermore, **Rasa Shastra**, a branch of alchemy in India, explored chemical processes for refining metals and preparing medicinal formulations. This knowledge significantly influenced Ayurveda and helped develop treatments for various diseases.

(C) Give characteristics of Indian philosophical and scientific traditions.

Ans. Indian philosophical and scientific traditions are deeply intertwined and characterized by their holistic, ethical, and interconnected approach to understanding the world.

1. **Integration of science and spirituality:** Indian traditions seamlessly blend empirical science with spirituality. For example, Ayurveda incorporates the concept of *panchamahabhutas* (five elements) and *tridoshas* (three doshas) into medical practice, emphasizing balance between body, mind, and nature.

2. **Emphasis on ethics and practicality:** Indian traditions prioritize ethical application of knowledge. Scientific advancements, such as sustainable agriculture and water conservation, were designed to ensure ecological balance and societal well-being.
 3. **Holistic thinking:** Indian knowledge systems view science, philosophy, and ethics as interconnected. This is evident in Vedic astronomy, where the study of celestial phenomena was linked to timekeeping, rituals, and agriculture.
 4. **Observation and reasoning:** Indian traditions emphasize empirical observation and logical reasoning. Schools like Nyaya and Vaisheshika promoted debate and logical inquiry as tools to uncover truth, influencing scientific disciplines like physics and mathematics.
- Overall, Indian traditions focus on the unity of knowledge, promoting harmony between humans, nature, and the cosmos.

Q.4 Answer the following (Any Two):
(Each question carries 3 marks)

(6 Marks)

(A) Describe traditional Indian perspectives on life sciences.

Ans. Traditional Indian life sciences, particularly Ayurveda, emphasize the harmonious coexistence of humans and nature. They are rooted in the concept that health and well-being depend on maintaining balance within the body and between individuals and their environment.

The concept of the **five elements** (*panchamahabhutas*)—earth, water, fire, air, and space—forms the foundation of Ayurveda. These elements are believed to combine into **three doshas** (Vata, Pitta, and Kapha), which govern the body's physiological functions. A balance of these doshas ensures good health, while an imbalance leads to disease.

Traditional practices also stress the importance of preventive care. For instance, maintaining a healthy diet, regular exercise, and yoga were recommended to keep the body in balance. Indian texts like the *Charaka Samhita* and *Sushruta Samhita* provide detailed insights into anatomy, surgery, and pharmacology.

Furthermore, Indian life sciences are deeply tied to ecological preservation. Ancient agricultural practices, such as planting medicinal herbs and crop rotation, reflected a profound respect for biodiversity and sustainability.

(B) Explain the significance of ancient knowledge.

Ans. Ancient knowledge systems have profoundly shaped modern civilization by providing the foundation for science, technology, philosophy, and sustainable practices.

Indian contributions to mathematics, such as the concept of **zero** and the **decimal system**, revolutionized numerical calculations

and laid the groundwork for modern computational techniques. Similarly, Aryabhata's work on planetary motion and Bhaskara II's advancements in algebra and trigonometry influenced global scientific thought.

In the realm of philosophy, texts like the **Bhagavad Gita** and **Upanishads** provided insights into ethics, leadership, and the purpose of life, which continue to inspire modern thinkers.

Environmentally, ancient Indian practices emphasized living in harmony with nature. Techniques like **rainwater harvesting**, **sacred groves**, and **organic farming** ensured ecological balance and sustainability. These practices are increasingly being revisited to address contemporary challenges like climate change.

Moreover, ancient knowledge teaches us to integrate ethics and scientific inquiry, ensuring that progress benefits humanity without harming nature.

(C) Describe Indian concepts related to electricity.

Ans. Ancient Indian texts, such as the *Yajurveda*, contain references to phenomena resembling electricity. For instance, descriptions of sparks caused by friction and the properties of lodestones indicate an early understanding of magnetic and electrical forces. Aryabhata's astronomical theories suggested that the Earth's magnetic properties influence planetary motion. This early knowledge laid the foundation for understanding electromagnetism and navigation using magnetic tools. Additionally, Indian artisans and alchemists explored the use of materials capable of generating sparks or static charges. Their practical applications, while not fully documented, likely influenced metallurgical processes and early experiments with conductive materials.

Q.5 Answer the following (Any One):

(6 Marks)

(Each question carries 6 marks)

(A) Write a note on any two ancient educational institutions.

Ans. Ancient India was home to some of the world's first and most renowned educational institutions, including **Takṣaśilā** and **Nālandā**, which attracted students and scholars from across the world.

- 1. Takṣaśilā:** Takṣaśilā, established around 1000 BCE in modern-day Pakistan, was a prominent centre of learning. It attracted students from India, Persia, Greece, and China. The curriculum was diverse, offering subjects like the Vedas, grammar, logic, medicine, astronomy, political science, and military training. Notable alumni include **Panini**, the Sanskrit grammarian who authored the *Ashtadhyayi*, and **Chanakya**, the economist and

strategist who mentored Chandragupta Maurya. Learning was informal, with no fixed classrooms or curriculum. Students studied under renowned teachers and engaged in debates and discussions to gain mastery over their chosen fields.

2. **Nālandā:** Nālandā, founded in the 5th century CE in Bihar, India, was one of the first residential universities in the world. It housed over 10,000 students and 2,000 teachers and had a vast library called **Dharmaganja**. Subjects taught included Buddhist philosophy, mathematics, medicine, and astronomy.

Scholars like **Aryabhata**, who developed the concept of zero and advanced theories on planetary motion, and **Xuanzang**, a Chinese monk, studied at Nālandā. The university symbolized interdisciplinary learning and became a hub for the exchange of knowledge between cultures.

- (B) **Describe ancient practices that promote environmental awareness.**

Ans. Ancient Indian practices emphasized harmony with nature, promoting sustainability and environmental conservation.

1. **Sacred groves:** Sacred groves were forested areas protected for spiritual and ecological purposes. These groves preserved biodiversity, served as habitats for wildlife, and maintained ecological balance. Religious practices ensured their protection, highlighting the connection between spirituality and environmental awareness.
2. **Water conservation:** Techniques like **step-wells**, **rainwater harvesting**, and **canal systems** were used to conserve and manage water resources. These systems ensured sustainable water use, particularly in arid regions, and supported agriculture.
3. **Sustainable agriculture:** Indian farmers practiced **crop rotation**, **natural fertilization**, and **organic farming**, maintaining soil fertility and reducing environmental degradation. Techniques like mixed cropping minimized pest damage and ensured ecological balance.
4. **Ethical principles:** The philosophy of **Ahimsa** (non-violence) extended to nature, promoting minimal harm to plants and animals. Festivals and rituals often involved tree planting and nature conservation, fostering environmental consciousness. These ancient practices are increasingly relevant today, offering solutions for tackling climate change and ecological degradation.

Practice Question Paper

B.Sc. & B.Ed. (ECS) I : Sem I

Introduction to Indian Knowledge System (IKS)

w.e.f. AY 2024-25 University Assessment (UA)

Time:

Total Marks: 30

Instructions:

- (1) All questions are compulsory.
- (2) Figures to right indicate full marks.

Q.1 Choose the correct alternative (MCQs): (6 Marks)

1. What is the primary goal of Indian Knowledge Systems (IKS)?
 - (a) Accumulating facts
 - (b) Achieving self-realization and liberation**
 - (c) Gaining material wealth
 - (d) Mastering Sciences and Arts
2. Which Indian philosopher is known for writing the "Ashtadhyayi"?
 - (a) Kautilya
 - (b) Panini**
 - (c) Aryabhatta
 - (d) Jivaka
3. What is the core concept of health in the Indian Knowledge System (IKS)?
 - (a) The absence of disease
 - (b) The balance between the body, mind, and environment**
 - (c) Physical fitness
 - (d) Mental clarity
4. Which Indian institution had the Dharmaganja library, one of the largest in the world?
 - (a) Takṣaśilā
 - (b) Nālandā**
 - (c) University of Nalanda
 - (d) Patanjali Academy
5. Which metal alloy was developed in India, known for its exceptional quality?
 - (a) Bronze
 - (b) Wootz steel**
 - (c) Cast iron
 - (d) Copper
6. Which traditional practice in India emphasizes the cultivation of medicinal plants for sustainability?
 - (a) Organic farming
 - (b) Ayurveda
 - (c) Sacred groves
 - (d) Both (b) and (c)**

(Q.1)

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Practice Question Paper

Q.2 Answer the following (Any Three)
(Each question carries 2 marks)

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(6 Marks)

- What is the significance of the concept of 'moksha' in Indian philosophy?
- Explain the use of natural dyes in ancient India and mention one important dye.
- What role did sacred groves play in environmental conservation in India?
- Which Indian mathematician introduced the concept of zero, and what was its significance?
- Describe the primary goal of Ayurveda and its approach to health.

Q.3 Answer the following (Any Two):

(6 Marks)

(Each question carries 3 marks)

- Discuss the distinctive features of Vedic and Non-vedic philosophical systems.
- How did Nālandā contribute to the spread of knowledge in ancient times?
- What are the contributions of Aryabhata to Mathematics and Science?

Q.4. Answer the following (Any Two):

(6 Marks)

(Each question carries 3 marks)

- How did ancient Indian scholars contribute to the understanding of motion and gravity? Discuss Brahmagupta's and Bhaskara II's ideas.
- Explain the principles of sustainability in ancient Indian agricultural practices.
- Discuss the significance of plants in Indian life sciences, including their medicinal and ecological roles.

Q.5 Answer the following (Any One):

(6 Marks)

- Discuss the role of Takṣaśilā and Nālandā in the spread of education and knowledge in ancient India.
- Explain the traditional Indian approach to environmental sustainability and its relevance today.

पुस्तक काळजीपूर्वक हाताळा